

1. This problem concerns the permutation $\mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 1 & 6 & 7 & 2 & 4 & 3 \end{pmatrix} \in S_7$

(a) Write μ as a product of disjoint cycles.

(b) Is μ even or odd?

(c) Find the order of μ .

(d) Find μ^{1000} .

2. Write the permutation $(215)(3142)^{-1}$ as a product of disjoint cycles.

3. Prove that if a permutation σ is even, then σ^{-1} is also even.